

Bhavya Kumar

 | LinkedIn | github.com/Bhavya2025 | bhavyakumar.ca

EDUCATION

University of Waterloo

Honours Bachelor of Mathematics (Co-op)

Waterloo, ON

Sept. 2025 – Present

- **Award:** President's Scholarship.
- **Relevant Coursework:** Data Types and Structures, Probability, Algorithm Design in C, Linear Algebra II, Calculus III, Functional Programming in Racket.

PROJECTS

Wyag - Git Version Control System - *Python*

May. 2026

- Engineered a fully functional Git client from scratch, implementing a Merkle DAG data structure and a content-addressed object store using SHA-1 hashing and zlib compression.
- Overcame binary parsing complexities by developing a custom parser for the Git index file to accurately manage the staging area, track file metadata, and detect worktree modifications.
- Resolved object identity discrepancies by writing custom lexicographical sorting algorithms for tree objects, ensuring perfect compatibility with the official Git CLI across 15 core commands.

Taskly - Kanban App with built-in AI Assistant - *React, Supabase, Vercel* | Live Website

Apr. 2026

- Diagnosed and solved a 60fps scroll regression caused by expensive CSS effects, rebuilding the HTTP layer with Node.js native HTTPS and passive event listeners to improve touch navigation performance.
- Eliminated cross-tab data inconsistencies by designing a centralized mutation pipeline with an undo/redo history and a 500ms-debounced Supabase sync, maintaining a single source of truth across guest and authenticated modes.
- Integrated Groq's DeepSeek-R1 70B to parse natural-language commands, utilizing a regex-fallback JSON extraction method to handle unpredictable LLM outputs and resolve fuzzy task/column queries.

Project Ghost - *Gemini 2.5 Flash, Python, Vultr*

Jan. 2026

- Developed a multimodal AI forensic code analysis engine that simultaneously parses source code and UI screenshots to generate plain-English debugging reports.
- Solved data-streaming bottlenecks by architecting a Python backend that bridges the Gemini vision model with ElevenLabs TTS, successfully deploying the optimized pipeline on Vultr for live hackathon demonstrations.

Model Self-Driving Car Prototype - *Arduino, Embedded C/C++*

2023

- Solved navigation deadlocks in an autonomous vehicle by writing embedded C/C++ control logic and calibrating ultrasonic detection thresholds through iterative real-world testing.

EXPERIENCE & LEADERSHIP

GDGI MUN

Head of IT

Apr. 2024 – Feb. 2025

- Directed IT logistics and media operations for a 2-day student-run Model UN conference, managing AV infrastructure, livestreaming, and registration systems across multiple committee rooms.

Clevered AI Fellowship

Remote

Student Fellow

Jan. 2024 – Feb. 2024

- Selected for a mentored AI program led by Oxford-affiliated researcher Ken Kahn; built a voice-activated accessibility prototype in Python using NLP-style intent matching to handle varied user commands.

TECHNICAL SKILLS

Languages: Python, C, C++, C#, JavaScript, Racket, SQL

Frameworks & Libraries: React, Node.js, Express, Supabase, dnd-kit, REST APIs

Tools: Git, GitHub, Vercel, VS Code, Arduino IDE

ACHIEVEMENTS

Bronze Medalist: TIMO International Math Olympiad

Finalist: Northeastern University London Essay Competition (Tech & Society)

Author: "School of Machine Learning" — blog simplifying ML concepts for beginners